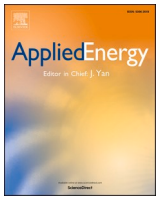
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Adaptive protection based on multi-agent systems for AC microgrids: A review

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HIGHLIGHTS

- Adaptive protection based on multi-agent systems (MAS) is a promising solution.
- Advances in MAS-based adaptive protection systems are systematically reviewed.
- MAS-based design and simulation for AC microgrid protection are included.
- A lack of contributions focused on real-time performance features is noted.
- MAS-based strategy's burden reduction of communications systems is also concluded.

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ABSTRACT

The bidirectional power flow, voltage/frequency dynamics, and reduced fault current observed in microgrids pose significant challenges to the protection of electrical systems. Particularly, the dynamic nature of microgrid-distributed energy generation requires protection schemes to adapt dynamically. Distributed protection strategies are commonly found in the literature, with adaptive protection based on multi-agent systems (MASs) being one of the most promising methods. This solution offers high autonomy, fault tolerance, and robustness against multiple fault types under various topology scenarios. This paper presents a systematic review of the current advances in MAS-based adaptive protection systems for AC microgrids. Furthermore, a discussion on the strategies implemented in MAS-based adaptive protection, the most used development frameworks for designing, developing, and testing MAS-based solutions for AC microgrid protection, and limiting factors for the widespread adoption are presented. Finally, open research problems and future research trends in MAS-based adaptive protection are also presented in this research.

1. Introduction

There is a widespread trend to increase the penetration of distributed energy resources (DER) into the power generation matrix, especially inverter-based DER (IBDER). However, the production capacity of IBDER is relatively low [1]. This low production capacity has stimulated the need to integrate various types of DER and corresponding loads as a microgrid. The definition of microgrid as per the International Council on Large Electrical Systems (CIGRE) is: 'Microgrids are electricity distribution systems that contain distributed energy loads and resources (such as distributed generators, storage devices, or controllable loads) that can be operated in a controlled and coordinated manner, either while connected to the main power grid or while in the island mode' [2]. Three major themes in microgrid (MG) protection challenges can be identified as: (i) bidirectional flow, (ii) reduced fault current capacity, and (iii) distinct voltage/frequency dynamics [3–5], as shown in Fig. 1. The bidirectional power flow between the main grid and the MG makes conventional MG protection techniques seem redundant or excessive [6]. Similarly, the different modes of operation and variety of capacities in the distributed generation (DG) contribute significantly differently to the fault current levels. During grid connected mode the main grid is in

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Nomenclature

ANN	Artificial neural networks
APS	Adaptive protection system
ATS	Automatic transfer switch
BESS	Battery energy storage system
BESSA	Battery energy storage system agent
BRAS	Back relay agent society
CBA	Circuit breaker agent
CPF	Cyber-physical framework
CI	Computer intelligence
CIGRE	International Council on Large Electrical Systems
CLR	Current-limiting reactor
DDS	Data distribution service
DER	Distributed energy resources
DG	Distributed generation
DGA	Distributed generation agent
DOCR	Directional overcurrent relays
ESS	Energy storage system
FCL	Fault current limiter
FLISR	Fault location, isolation, and service restoration
FRAS	Forward relay agent society
HIF	High-impedance fault
HIL	Hardware-in-the-loop
HPS	Hierarchical protection scheme
IBDER	Inverter-based distributed energy resources
IBDG	Inverter-based distributed generators
IED	Intelligent electronic device
LA	Load agent
MAS	Multi-agent systems
MG	Microgrid
ML	Machine learning
PCC	Point of common coupling
PMU	Phasor measurement unit
PV	Photovoltaic
RA	Relay agent
RAG	Relay agent group
SBDG	Synchronous-based distributed generators
SCADA	Supervisory control and data acquisition
VLAN	Virtual local area network

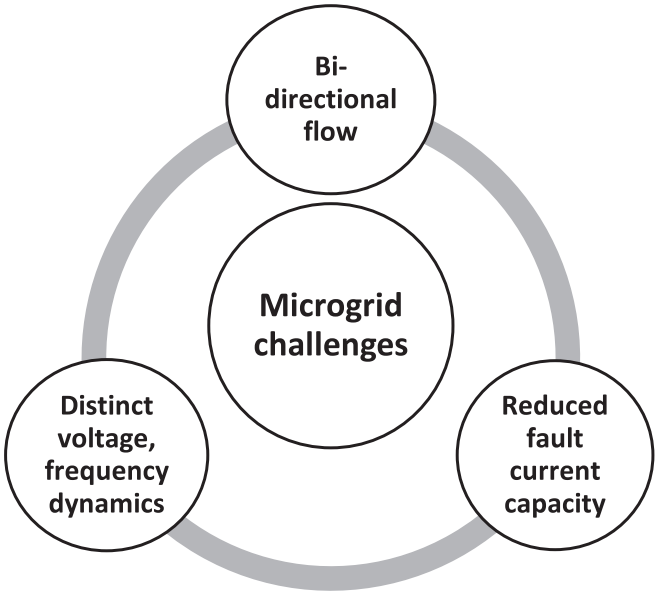


Fig. 1. Main challenges in MG protection.

charge of supplying the required power demand to the loads, providing voltage and frequency stability within the microgrid. Under islanded-mode conditions, this challenging burden falls to the limited microgrid resources [4,5,7].

As proposed in [8], the presence of DGs that contribute to the fault currents can introduce unpredictable operating conditions in the existing protective devices that disrupt the coordination of the electrical protection systems, leading to a loss of functionality. Moreover, Abasi et al. [9] affirmed that the existence of renewable energy-based MGs with variable bidirectional power flow leads to relay overreach and underreach at the other two terminals. Consequently, the traditional protection techniques, such as differential, overcurrent, and distance protection, are insufficient to tackle the current challenges [10]. Moreover, network configurations no longer exhibit radial structures with a unidirectional power flow and, consequently, MG protection strategies have been introduced in the literature to address the technical challenges posed by the increased DG. Among these protection methods are differential protection, voltage-based protection, current-based protection, impedance-based protection, distance protection, and directional protection [11]. For this reason, there is a current need for protection techniques that can adapt to the dynamic changes in the power distribution network with modern interconnected MGs [12,13]. Various fields of interest can be identified, being conducted toward the development of such adaptive protection schemes [14]. One option is the use of the multi-agent system (MAS) [15]. MAS-based adaptive protection involves the use of autonomous or semi-autonomous intelligent computer systems of either hardware and/or software solutions, which possess critical decision-making abilities based on the local or global information that, by working collectively, can achieve the designed goals [16]. In [10], an MAS was applied using the grid-connected devices, such as the loads, DGs, and relays as agents. The authors noted that these agents were able to communicate with each other the collected information of power systems and its equipment. They perform the necessary computations to decide the actions that best preserve the protection system coordination. Therefore, MAS-based solutions combined with traditional protection techniques have high potential for addressing these issues. Nevertheless, the communication architecture deployed for the MAS based on adaptive MG protection requires certain features to maintain optimal performance, robustness, low latency, and ultrahigh availability, as indicated in [17]. Based on the analysis of other recent review contributions [1,9,16,17], a lack of comprehensive coverage regarding the use of MAS-based protection and its various approaches is noticeable.

This paper aims to provide a review of the challenges, current advances, and future trends in MG adaptive protection schemes based on MAS, as well as the technologies used to develop them. The main contributions are: (i) an examination of the current conventional and adaptive MG protection approaches; (ii) a literature review of the current trends in microgrid adaptive protection schemes based on MAS and the devices used for their implementation in recent contributions; and (iii) an overview of the development tools and communication infrastructure for microgrid MAS-based adaptive protection schemes.

The remainder of this paper is organized as follows. Section 2 discusses the applied methodology. Section 3 covers the solutions proposed in the previous literature, highlighting the adaptive protection. In Section 4, the MAS solutions for adaptive protection are described, followed by a detailed discussion in Section 5. Finally, Sections 6 and 7 complete this review by providing future research directions and conclusions respectively.

2. Methodology

This review was developed based on the following methodology. (i) Data recollection: A detailed search was performed on four principal scholarly databases: Scopus, IEEE, Web of Science, and Google Scholar. These databases were selected because of the large number of publications they had access to. They have a broader scope on the subject than the data that could be obtained from individual sources. Cutting-edge research from the last five years was selected, while more foundational works were selected from the earlier periods. The keywords "microgrid", "smart grid", "adaptive protection", "multi-agent systems", "energy storage", "microgrid communications", "IEC 61850", among others, were used. (ii) Data extraction and sorting: With the assistance of a reference manager, the collected literature was organized based on the titles, keywords, and abstracts. Subsequently, the contributions of these studies were identified and classified. Non-relevant contributions or studies outside the scope of this revision were discarded. (iii) Subtopic selection and assignment: Relevant literature was analyzed to identify the underlying relationships among the contributions. Additionally, the subtopics used in this study were identified. (iv) Information analysis: The collected data were subjected to critical analysis, resulting in the conclusions. Initially, the authors analyzed the contributions of the relevant reviews, and subsequently, the most recent primary research papers on each subtopic were identified.

3. MG protection approaches

There are underlying conditions that affect the nature of the MG protection strategy. These conditions are as follows:

- **Topology:** MG systems can be single-bus, multi-bus, ring/mesh, looped, or mixed [18,19]. The topology could change as the operational mode changes from grid-connected to islanded mode, affecting the power and current flow direction [20]. Some countries have standardized anti-islanding methods to prevent unintentional disconnections from the primary grid [21].
- **Fault type:** MG can experience multiple faults, such as voltage sag faults, single and three-phase faults, low and high-impedance faults, short circuit faults, and symmetrical and unsymmetrical faults [19].
- **Type of MG:** Fundamentally, microgrids can be classified as AC, DC, and hybrid. However, they can be further categorized depending on the DERs size, scenario, and operational mode [22]. It is important to note that AC and DC protection equipment have different ratings when engineering a protection strategy [19]. In [23], the authors presented a recent MG classification as a cyber-physical framework (CPF), being defined as a heterogeneous network that combines two different homogeneous networks, i.e. power (physical) network and cyber network. Therefore, and based on the CPF concept, the control structure can be also divided into two layers: the cyber layer and physical layer.
- **Type of DERs:** The DERs in a microgrid can be either a traditional synchronous machine based or IBDER. Synchronous generators can contribute a much higher fault current level than IBDER, consequently affecting the protection scheme [21].

These factors influence the overall MG protection approach by affecting the direction of the current flow, the fault current levels, and voltage/frequency dynamics. These conditions determine the MG stability and operation conditions. Indeed, the presence of MGs with varying power generation levels significantly alters the network's short-circuit capacity, which directly impacts the settings and coordination of distance relays, as well as fault location algorithms. Additionally, furthering the application of internationally accepted standards such as IEEE 1547 and IEEE 2030 can significantly enhance the reliability of these power systems [24]. These standards set the criteria and interoperability requirements for interfacing smart grid technologies and power systems.

3.1. Non-adaptive protection

Overcurrent relays and fuses have been conventionally used as the solution of choice for radial power distribution systems primarily because of their simplicity [25], and due to the short-circuit current level being often considerably different after changes in topology [19]. Consequently, because of the reduction in the fault current level during the islanded mode, traditional overcurrent and overload protection become inadequate because their settings are higher than the required threshold [20]. To overcome the difficulties caused by the bidirectional power flow, directional overcurrent relays (DOCR), which are significantly more sensitive than the nondirectional overcurrent relays, are used in the MGs [26]. Overcurrent relays can also address symmetrical and unsymmetrical faults [27]. Differential protection schemes provide faster responses than the traditional overcurrent protection schemes, although they require measurements from remote terminals [26]. These schemes are widely recognized for their high sensitivity, fast response, and reliability, being suitable for both islanded and grid-connected modes [28]. It is also reported that high-impedance faults can also be handled with differential protection schemes [27]. Traditional differential protection schemes encounter challenges related to data synchronization mismatches [29]. The distance protection, which is an impedance-based scheme, is considered suitable for transmission lines where sensitivity and selectivity issues are significant concerns [30]. However, it has the disadvantages of being nondirectional and vulnerable to power swings and heavy loads [28]. In [31], an intelligent distance relay is proposed based on an Intelligent Electronic Device (IED) operating on the IEC61850 communication network. This relay overcame a relevant limitation found in traditional distance relays, which fail to provide instantaneous protection for the entire line. In [32], a designed distance protection scheme was configured to effectively safeguard against both solid faults and faults with significant resistance, regardless of type, even under high DG penetration and addressing weak infeed conditions. More specifically, the impact of high PV integration into the reliability of phase and ground distance elements, monitored by negative-sequence directional elements, was discussed in [33]. More recently, a single-ended intelligent protection method, based on a combined linear trend approach, was developed and evaluated in [34] for detecting faults, determining fault locations, and classifying faults on lines connected to high integration of PV systems. An impedance-based device's accuracy may be affected by DGs located between the fault and the protection device, transients, harmonics and fault resistance can influence the impedance measurement [35]. As presented in [36], the fundamental concept of voltage protection is based on the under- and over-voltage events. However, the difficulty in identifying both transient and permanent faults limits the application of this scheme because of the frequent occurrence of undervoltage and overvoltage conditions in power systems. Fig. 2 presents the various nonadaptive protection schemes mentioned in the literature. As can be seen, auxiliary devices are also part of the non-adaptive solutions. Fault current limiter (FCL), energy storage system (ESS) solutions, and automatic transfer switches are also included in this category. These components are not thoroughly considered part of the MG protection system in the current relevant literature, even though they can play an even more applicable role in providing additional support to the most used protection devices. FCLs are a set of devices installed near the point of common coupling (PCC) to control and limit fault currents within an MG for both internal and external faults. Under normal operating conditions, the FCL maintains a minimum impedance setting to prevent undesirable voltage drops and losses. However, under fault conditions, it dynamically switches to the maximum impedance to limit the magnitude of the fault current [36]. To further mitigate the effect of the DERs on the MGs, a high-inductance current-limiting reactor (CLR) is inserted to prevent rapid increases in fault current. This reactor effectively blocks the spread of fault current to

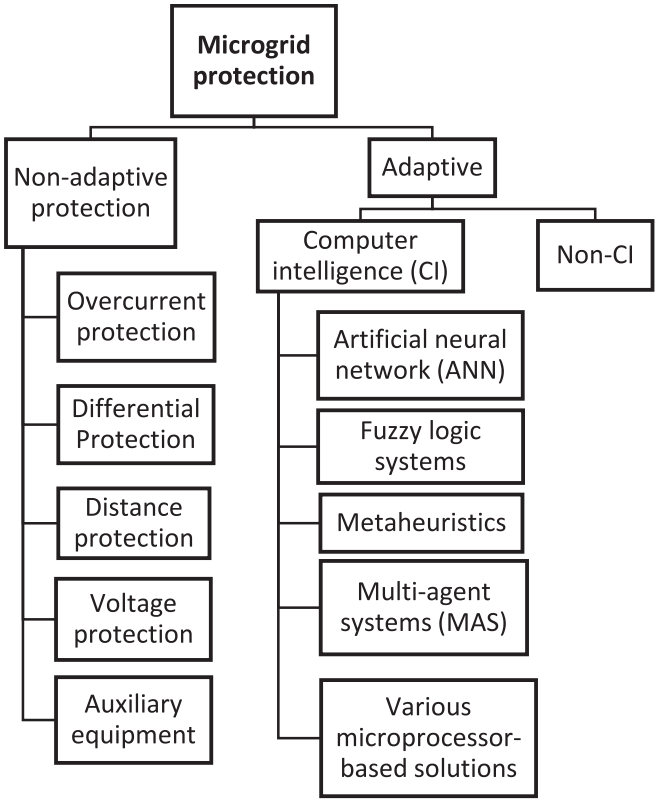


Fig. 2. Microgrid protection approaches.

the unaffected parts of the system, thereby minimizing power disruptions.

Within a MG, the inclusion of an ESS has become increasingly crucial since it can play a significant role in guaranteeing a dependable and steady power supply for the local loads [18,19]. Multiple applications of ESSs have been identified, such as supply and demand balancing, grid and RES buffering, market operations, scheduling, peak shaving, power quality enhancement, emission control, service efficiency, emergency, and black start support [37]. Some researchers have proposed schemes, although few, in which the ESS is applied as an active part of the MG protection mechanism, as mentioned in [38–40]. In [41], the authors have discussed the ESS (flywheels, supercapacitors, and SMES) being interfaced with higher-rating inverters to improve the fault current levels. The findings presented in [39] demonstrate that a supercapacitor already existing in the test system effectively increases the fault current to trigger a protection relay. Habib et al. [42] proposed the implementation of a lithium-ion battery to sustain operation during communication failures and augment fault current levels when the system fails to detect the transition into the islanded mode. This study focuses on reducing the reliance on mode detection for proper protection operations. The proposed protection scheme employs messaging based on the IEC 61850, with the data distribution service (DDS) middleware serving as a link between the hardware and software environments. Habib et al. [43] proposed using supercapacitors to augment the efficacy of the adaptive protection schemes during communication failures. In this scenario, supercapacitors are employed to introduce fault currents into the AC side of the MG, consequently elevating the fault current level and mitigating the necessity for the mode of operation commands in the event of communication issues. Fawzy et al. [44] assembled a co-simulation platform based on a multiagent system and battery attached to an MG. Resultingly, the settings of the protective relays need not be changed in the different modes. Therefore, they were not adaptive, and the protection scheme cost was low because the battery was used to feed the pulsed load. The idea behind these studies is that the ESS, as a component of the protection system, contributes to the introduction of an additional fault current to the power distribution grid to enhance its resiliency in the islanded mode. This concept can be found in other studies published in the previous decades [41,45].

Reclosers play a vital role in the automatic reconnection and reenergization of power lines. They can detect and interrupt overcurrents, thereby preventing the circuit from remaining open owing to temporary disturbances [46]. However, in low-inertia MGs, this scenario can become critical because reclosing under a permanent fault can cause the MG to continue disturbances. Postfault oscillations in DGs can lead to system instability. Additionally, issues related to relay-fuse selectivity and miscoordination arise because of the introduction of fault currents from the DGs. Chandra et al. [36] presented a solution to this problem by modifying the operating parameters of the reclosers. An automatic transfer switch (ATS) is employed to reconfigure the network and ensure uninterrupted power supply to the unaffected zone [46]. DGs equipped with backup generation employ an ATS to restore power to all the loads during faults using a backup source. However, it is crucial to consider the effect of latency before transferring the DGs [47]. After clearing the fault, the ATS switches back to operation on the grid supply and issues a shutdown command to the backup DG. It is essential for all the reclosers or breakers to complete their reclosing sequences before the ATS initiates its operation [36]. Barra et al. [4] examined the potential option of disconnecting the DGs during the occurrence of a fault. DGs equipped with backup generation employ an ATS to restore power to all the loads during faults by utilizing a backup source. However, it is crucial to consider the impact of latency before the DG transfer [47]. Once the fault is resolved, the ATS switches back to the grid supply operation and issues a command to the backup DG to shut it down. Notably, all the breakers and/or reclosers should complete their sequences prior to the ATS [16]. However, it has several drawbacks, such as voltage spikes owing to the removal of the reactive power contribution from the DGs. Notably, the strategy of limiting the DG capacity is not advisable, as it is critical to extract the most value from the DGs in terms of energy production and CO_2 emission reduction. Additionally, as argued in this contribution, adding more protection and FCL devices is costly for the utility and DG owners.

3.2. *Adaptive protection*

Phadke et al. [48] introduced the concept of adaptive protection as a strategy to enable and optimize various protection functions to align better with the changing conditions of a dynamic power system. This involves adjusting the protection settings and methodologies to enhance their suitability and effectiveness in response to the variable system conditions. The adaptation of relay settings based on the system conditions can be traced back to 1967 [49]. Therefore, this is not a new idea, although various studies have proposed novel approaches for its efficient implementation. The adaptive protection of the MG systems is being continuously developed. There is ongoing research to properly optimize this type of protection as well as to overcome the possible drawbacks [15]. One of these limitations is the dependence of power systems on a highly effective and reliable communication infrastructure. Therefore, decentralized protection capability is an important feature for modern smart microgrids. In [50,51] and adaptive protection scheme was implemented through a centralized supervisory control and data acquisition (SCADA) system, which as able to provide improved coordination, selectivity and sensitivity of protection relays, but completely dependent on central processor and does not mention how it would react during a communication failure. Although SCADA are simpler to commission and validate than agent systems, which are addressed in the next section. Additionally, adaptive protection is a complex system compared to the ordinary power system protection, requiring more time for the system to work through the fault. Hence, a crucial aspect of adaptive protection is its requirement to operate as closely as possible in real time [15].

The literature review indicates that the adaptive protection for MGs research has focused on exploring computer intelligence (CI) techniques, including the artificial neural networks (ANN), as in [52]. In this paper the authors implemented ANN combined with directional relays solving the drawbacks of conventional protection schemes. In other works, researchers have applied deep learning techniques to achieve high accuracy in fault detection and fault location [53–55]. These are very effective strategies but computationally very expensive. Other approaches include fuzzy logic systems, metaheuristics, multi-agent systems, as discussed in Section 4, and other alternative intelligent approaches. These CI techniques have been extensively studied and applied in the context of adaptive protection of the MGs [18]. A hierarchical diagram of the different approaches can be found in Fig. 2. Alternatively, other studies have proposed adaptive protection features without the use of the CI methods, mostly through offline calculations of the parameter settings. In [56], an example was leveraged on the quadrature and zero-sequence components of the fault current considering the impact of the reactance divided by the resistance (X/R ratio) of the DGs. Additionally, in [57], the state of the electrical devices was monitored, and the operating mode was identified to ultimately manage the adjustment of the intelligent electronic devices (IEDs) and consequently to achieve suitable and reliable MG protection. A genetic algorithm was used in [58] to determine the most optimal photovoltaic (PV) sources penetration level to maintain protection system coordination. A non-communication solution was used because it was not necessary to adjust the protection device settings. PV sources were only considered in this study, thereby limiting the amount of DG required to maintain coordination. If there were any changes in the PV penetration level, offline calculations would have to be recomputed. Barra et al. [4] also identified that a significant portion of the existing studies on adaptive protection primarily focused on overcurrent protection strategies. Some studies combined overcurrent protection with distance protection, such as in [59], though the distance protection would carry the drawbacks with impedance-based solutions, as mentioned in Section 3.1. Therefore, there is potential to explore the application of other protection philosophies, including hybrid models. Most studies evaluated the performance of the suggested methods only for three-phase balanced faults. However, there is a growing need to consider the MGs and distribution systems (DSs) that incorporate both the inverter-based distributed generators (IBDG) and synchronous-based distributed generators (SBDG). The authors concluded that future research should consider the presence of ESSs when proposing and evaluating new adaptive protection schemes for the MGs. A summary of the recent contributions to adaptive protection is presented in Table 1.

4. MAS-based adaptive protection

In the context of adaptive protection systems for MGs, the term ‘agent’ refers to a software or hardware component that operates autonomously to adapt to the changes in its surrounding environment [76]. Agents utilize data received from the network as well as the information from other agents to make informed decisions and execute the predefined functions [77]. These agents are typically installed on intelligent electronic devices (IEDs). They possess the computational capabilities to run the agent algorithms and require a dedicated infrastructure that encompasses the communication links and protocols. Through these communication protocols, agents can transmit data across the communication channels of a MAS-based hierarchical protection scheme (HPS) for distribution power systems while adhering to specific policies and simultaneously receiving data from other agents. The communication protocol was designed to facilitate the exchange of messages between the two agents, enabling them to accomplish specific tasks [77]. Effective decision making by agents requires collaborative communication to obtain the network conditions. A MAS consists of two or more agents working together in an environment, typically organized in a hierarchical structure with centralized, distributed, or hybrid relationship architectures. Chandra et al. [36] noted the potential use of a SCADA system assisted by multiple agents for wide-coverage area monitoring and protection. This integrated approach offers both primary and backup protection while considering the DG effects outages on the protection scheme. The reviewed systems facilitate simultaneous updating of the relay and DG controller settings to ensure coordinated and effective protection. The implementation of time-synchronized measurements enhances fault detection and isolation accuracy. Indeed, and mainly due to the frequent changes in MG topology, relays with fixed settings can fail under certain circumstances, such as heavy distribution loads, high levels of DG penetration, islanded or grid-connected modes, and radial or meshed topologies [78]. The authors observed that multi-agent communication played a crucial role in power restoration for loads and in improving the system reliability. However, this centralized technique relies heavily on communication and requires consideration of the latency for real-time operation. Although this reduces the cost of voltage transducers by relying solely on the current measurements, the addition of phasor measurement units (PMUs) and advanced communication infrastructure increases the overall cost.

MAS-based technology can be effectively combined with the existing data assessment tools for advanced power system diagnosis [20]. Considering the availability of multiple platforms, the selection of a MAS platform is crucial for ensuring compatibility, robustness, and durability during online implementation. Given the peer-to-peer nature of the agent systems, security has emerged as an important consideration. Notable concerns encompassing mobility, security, data standards, and intelligent agent design were identified in [20], necessitating comprehensive resolution. Moreover, the detection of high-impedance faults within the MAS-based solutions was emphasized in [1]. Similarly, the loss of control in the event of a communication failure is also a significant concern associated with the MAS-based approaches, as mentioned in the literature.

Cybersecurity is one of the most significant issues encountered by the agent systems. Owing to the presence of different layers in the MASs, as shown in Fig. 3, proper identification is required for online operations. The figure shows the various types of agents, such as the relay agent (RA), DG agent (DGA), load agent (LA), and BESS agent (BESSA). Other types of agents can also be implemented in the MAS-based solutions. Sarangi et al. [79] argued that the main problems associated with a multiagent-based protection strategy are the inability to detect high-impedance faults (HIF), service restoration issues in the network, and inadequate control during communication system failure.

In [8], the possibility of a comprehensive adaptive scheme encompassing both the local and backup protections was discussed. The authors acknowledge that implementing such a scheme, in which each relay agent is interconnected with others, would be challenging for practical applications in large MGs owing to the high computational and communication requirements. Although multi-agent-based protection schemes offer advantages over the MG central protection unit (MCPU)-based schemes, such as eliminating single-point failures and eliminating the need for dedicated communication systems, they may not be economically viable or beneficial for smaller MG structures. It was revealed in [10] that hierarchical, hybrid, and decentralized MAS architectures are commonly employed in research activities. The authors also noted limited research on utilizing the MASs for power system protection and fault location, isolation, and service restoration (FLISR), although the MAS techniques can be applied effectively to suitable architectures. Furthermore, Shobole et al. [10] emphasized that communication between the agents and within the MAS poses notable challenges for industrial implementation. They identified the need for wider adoption of the MASs and system integrity protection schemes as the research gaps. Furthermore, real-time execution is recognized as an area requiring further investigation. Bridging the gap between the MAS techniques and industrial products is an area for future research. The authors outlined several challenges associated with MAS, including selecting the appropriate architecture, ensuring communication

Table 1
Summary of recent adaptive protection contributions.

Ref.	Year	Approach	Architecture	Mode	Fault types	Fault detection strategy	Testing/ simulation	DG type	Comm.	Advantages	Drawbacks
[53]	2024	CI	DT	GC Is	LG, LL, LLG, LLLG	DWT	MATLAB DigSILENT	SBDG, IBDG	No	– High accuracy in fault detection and fault location – No communications needed	– Computation intensive neural network training required
[55]	2023	CI	(–)	Is	LG, LL, LLG, LLLG	ANN	MATLAB	(–)	No	– No communications needed – fault detection, classification, and locating	– Computation intensive neural network training required
[60]	2022	OT	CT	GC	(–)	OC	MATLAB	SBDG, ASBDG	Yes	– There is no operating time delay, and it is relatively simple	– Failure of the central controller affects the whole system
[58]	2022	CI	DT	(–)	LLL	OC	MATLAB DigSILENT	IBDG	No	– No adjustments needed among protection devices	– Only PV DG is considered and limits the generation to maintain CO
[61]	2022	CI	(–)	Is	LG, LLL	OC	MATLAB	IBDG	No	– Adaptability to FCL	– Only IBDG is considered
[62]	2022	CI	DT	GC Is	LG	OC	MATLAB ETAP	(–)	No	– Low tripping time	– Real-world performance evaluation needed
[63]	2021	CI	CT	GC Is	LG, LL, LLG, LLLG, HIF	OC	RSCAD/ HIL	All	Yes	– Reduced computation burden – No damage MG in critical situations	– No variety of network configurations – Non-standard relay curves not considered
[64]	2021	OT	CT	(–)	LLL	OC	PSCAD EMTDC	SBDG, IBDG	Yes	– Employs ESS as part of protection strategy	– Various types of DG not tested
[65]	2021	CI	(–)	GC Is	LLL	OC	MATLAB DigSILENT	All	(–)	– Reduced coordination times – Consideration of non-standard characteristic	– SBDG stability constraints not considered
[66]	2020	CI	Hy	GC Is	LLL	OC	DigSILENT	SBDG, IBDG	Yes	– Decentralized communication capability	– Potentially limited by number settings groups
[59]	2019	CI	CT	GC Is	LLL	OC, Dist	DigSILENT	SBDG	(–)	– Improves resiliency – Low cost	– Time consuming learning process – IBDG not considered
[67]	2019	CI	CT	GC Is	LLL	OC	OT	All	Yes	– Real-time operation	– Heavily dependent on communication link
[68]	2019	OT	DT	GC Is	LLL	OC	Simulink	SBDG	Yes	– No communication system needed – Fast fault clearance	– It might not work for systems with impedance
[56]	2018	OT	DT	GC Is	All	OC	Simulink	All	Yes	– Reduced dependence on central management via fractionalization of protection	– Dependent on good communications – Extensive calculations are needed
[69]	2018	CI	CT	GC	(–)	OC	OT	SBDG	Yes	– Clustering selects # of clusters as same as relay settings groups	– Optimized configuration not guaranteed
[70]	2018	OT	CT	GC	(–)	OC	DigSILENT	All	Yes	– Considers power grid uncertainties that could affect coordination	– Cannot detect open conductor faults – Islanded mode not considered; only for grid-connected mode
[71]	2018	OT	CT	GC	LLL, LLG	OC	DigSILENT	IBDG	Yes	– It only requires a voltage profile to adapt protection settings	– Falling voltage profile is weakened with a change in topology and an increase in impedance
[72]	2017	CI	(–)	GC Is	(–)	OC	OT	SBDG	(–)	– Specially designed for topology changes	– IBDG not considered – Optimization is high time-consuming
[50]	2017	CI	CT	GC Is	(–)	OC	OT	All	Yes	– Automatic coordination – Sensitivity improvement	– Heavily dependent on communication link
[73]	2017	CI	CT	GC	LLL	OC	RSCAD	SBDG	Yes	– Real-time calculation of relay settings	– HIL is suitable as a test bed for development but expensive for real-world application

(*continued on next page*)

Table 1 (*continued*)

Ref.	Year	Approach	Architecture	Mode	Fault types	Fault detection strategy	Testing/ simulation	DG type	Comm.	Advantages	Drawbacks
[74]	2017	CI	CT	GC Is	LLL	OC	ETAP	SBDG	Yes	– Robust against selectivity changes due to DG interconnection	– Influence of fault current not considered
[75]	2017	OT	CT	GC Is	LLL, LG	OC	RSCAD	All	Yes	– All fault types, including HIF, are considered	– Suitable for systems in GC or IS, but not in both

Legend:
– Approach: Computer intelligence (CI)
– Challenge Focus: Coordination (CO)
– Architecture: Distributed (DT), Centralized (CT), Hybrid (Hy)
– Mode: Grid-connected (GC), Islanded (Is)
– Fault types: Single-phase (L), single-phase (L), Three-phase (LLL), Two-phase (LL), to ground (−G), High impedance fault (HIF)
– Fault detection strategy: Overcurrent (OC), Distance relay (Dist), Discrete wavelet transformer (DWT)
– DG types: Synchronous-based distributed generation (SBDG), Asynchronous-based distributed generation (ASBDG), Inverter-based distributed generation (IBDG), including any DG.
– Hardware-in-the-Loop (HIL), Other (OT), Not clearly specified (−)

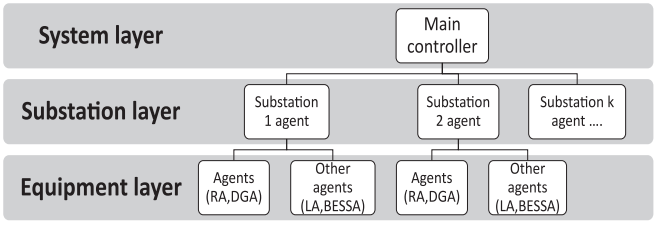


Fig. 3. MAS hierarchical layers.

standardization and reliability, facilitating coherent decision-making, tolerating multi-point failures, and addressing the lack of practical application experience. They argued that the interoperability requirements of the MASs could be effectively addressed using the IEC 61850 standard protocols. In [13], these findings were corroborated with the main issue of the MASs identified as increased complexity and cost of communication protocols as the number of agents increased. In [27], the temporary circumstances and different operation modes of the MGs were not adequately considered. The proposed schemes were found to be time-consuming and imposed a significant burden on the communication systems. The authors also highlighted the potential threat to cybersecurity associated with using the TCP/IP for communication between the MASs and external devices/software. Moreover, they suggested the need for improvements in the response time. Table 2 summarizes recent review contributions focusing on the MAS solutions.

An adaptive MAS-based approach was proposed in [80] for distribution networks (DNs) with wind turbines. The lowest layer of the MAS consists of distributed controller agents and relay agents, namely, the DG agent (DGA), relay agent group (RAG), and load agent (LA). The RAGs are further classified as the back relay agent society (BRAS) and forward relay agent society (FRAS). The loads and DGAs are organized within the load agent society (LAS) and DG agent society (DGAs), respectively. The control center assumes a coordinating role by executing various tasks, including the system-wide information gathering, network structure identification, operation mode identification, agent society formation, and mapping relationship revision. The RAG and DAG implement an adaptive protection strategy, in which the DGA continuously adapts the reference current of the grid-side converter to mitigate the injection of fault current under fault scenarios. Rahman et al. [81] introduced a multi-agent framework for distribution feeder protection systems. In this framework, each agent fulfills the protection relay requirements through information exchange. The proposed scheme demonstrated high reliability and accuracy with a response time

Table 2
Summary of reviews that consider MAS solutions.

Ref.	Year	Focus	New perspectives for MAS solutions
[21]	2023	– AC MG protection – Feature extraction in AI-based methods – Protection methodologies	– Real-time experimental investigation – Cybersecurity resiliency – Potential loss of coordination between the primary and backup relays
[10]	2021	– MAS in smart grid protection – FLISR – IEC61850	– More MAS-based techniques for power system applications – MAS research works on system Integrity Protection Schemes – Further research MAS suitable for Real-time control
[36]	2021	– AC Fault's current behavior – Grounding – AC MG protection – Standards – Communications	– Real-time or near-real-time operation – Communication and cost burden reduction
[1]	2021	– MG Frameworks – AC and DC protection	– Identifying high-impedance fault – Improve robustness against comms loss. – Cybersecurity in MAS-based systems
[8]	2021	– Literature review – Protection algorithms comparisons – IBDG	– Communication techniques and communication protocols to make schemes more reliable and economical. – Efficient islanding detection technique
[27]	2021	– Adaptive protection	– Communication techniques and communication protocols to make schemes more reliable and economical. – Real-time or near-real-time operation – Communication and cost burden reduction
[20]	2020	– Review survey articles – AC MG adaptive protection	– Mobility, Security, Data Standards, and Intelligent agent design – Cybersecurity
[79]	2019	– Hybrid AC/DC MG protection	– Detect the HIF – Service restoration issue in the grid – Better handle in communications failure
[13]	2019	– MG Adaptive protection with DG	– Scalability with the increased number of agents

of 18–22 ms for fault clearance and power restoration. The authors highlighted that the scheme was capable of handling faults even during primary protection system and communication link failures owing to the backup protection provided by adjacent RAs. Shirazi et al. [82] proposed the MAS-based structures for monitoring, fault identification, and protection. Each structure consists of agents assigned distinct roles, such as information gathering, database management, state diagnosis, decision making, execution, rule base, and communication. The information-gathering module is responsible for acquiring data on the voltage, power, and current parameters. Subsequently, the collected information was classified and analyzed using the database module. The authors utilized MATLAB® and NetLogo® to validate their proposals, focusing on single-fault scenarios and assuming a predefined fault location for the simulations. No location algorithm was used in this study. Nevertheless, and in accordance with these recent contributions, the inherent delay or failure of communication system must not affect the vital role of fault clearing. In addition, a reduction in communication network bandwidth would allow for faster detection and isolation of faults, resulting in a more cost-effective scheme for reliable and multipurpose operation of MAS-based protection schemes. Recent contributions affirmed that the requirements for the operation of power systems with high inverter-based resources after the fault clearance were not consistent across most regulatory frameworks, such as the NERC recommendations or the IEEE Standard 2800–2022 [83]. In addition, the adaptive protection schemes recently demonstrated potential for fault detection and isolation, contributing to reliable and secure MG operation. Aghdam et al. [84] developed a MAS-based variable tripping time differential protection scheme aimed to protect MGs in both standalone and grid-connected modes, considering the HIF effects. The proposed scheme also included fast fault clearance by coordinating agents with the lateral systems.

A bi-level protection scheme was proposed in [85], consisting of relay and generation agents. The scheme operates at two levels to ensure reliable and coordinated protection performance. At the first level, the relay agents control their own operations as well as the operation of neighboring agents. This control was based on their interrelationships to prevent malfunctions during fault conditions. At the second level, the generation agents manage power distribution within their respective areas by considering the relationships between them. This management strategy aims to prevent potential malfunctions in the overall protection system. Satuyeva et al. [86] presented a Q-learning technique combined with an MAS, which was used to make decisions regarding the faults in a distributed architecture. Circuit breaker agents (CBAs) were used in a master-slave MAS with the aim of applying them to the distribution system protection frameworks. The MG was modeled in Matlab®, and the MAS agents were implemented using the JADE platform. An MAS in conjunction with machine learning (ML) was suggested in [87] for fault diagnosis, protection coordination, and relay settings to achieve adaptive protection. The authors addressed both the islanded and grid-connected modes in their study. Following a fault event, the agents engage in communication and coordination to accurately locate and identify faults. The simulation results confirm that the proposed MAS algorithm can provide primary and backup protection for autonomous and grid-connected MGs. AnyLogic® simulation software was used to validate the agent interactions. The use of overcurrent relays was proposed in [88] as the agents equipped with IED capabilities to achieve self-adaptation to the changes in the DG penetration and system topology. This proposal uses the IEC 61850 protocol for peer-to-peer communication. The agents were segmented into the virtual local area networks (VLANs), for the radial and loop topologies. The relays detected faults based on their characteristic curves. The MAS algorithm was used to detect faults occurring within the VLAN. The authors concluded that the proposed scheme was not dependent on the location and penetration level of the DGs or on the location and type of faults. However, large MGs were not considered, being an important drawback of this proposal.

A self-healing system integrated with an adaptive protection system (APS) was proposed in [89]. This approach was combined with the MAS techniques to address the miscoordination issues caused by the increased penetration of the DGs. In [16], an HPS with the MAS was proposed. In essence, it controls the power produced by the DGs to manage the fault current that would be handled by the HPS primary protection. In this case, only the DGs act as the agents. The authors highlighted that the approach is not susceptible to communication link failures among the protection levels. A combined adaptive and multi-agent protection scheme was discussed in [12] based on a superposition theorem-based formula. After clearing a fault, the relay group settings are updated to adapt to the changes in the MG topology. In this study, single- and multi-event faults were considered in the proposed scheme. Calculations were performed offline for single-event faults and subsequently processed in the equipment layer. Multi-event faults were computed online and processed in the higher layers. The authors suggested that online calculations could be excellent for continuing research on MG protection issues. A fully distributed MAS-based APS using numerical DOCRs as adaptive protection agents was proposed in [90]. The MAS-based adaptive backup-protection scheme was used as a supplement using the settings group of the backup relays. Daryani et al. [91] described a method based on a relay as an MAS agent in conjunction with the DGs, current transformers (CTs), points of common coupling (PCC), and voltage transformers (VTs). The MAS architecture comprises three primary subagents: measurement, operation, and communication. The operation subagent includes key components, such as processing, direction identification, mapping detection, and memory functions. The relay, functioning as a MAS agent, exhibited substantial autonomy in detecting the network configuration, dynamically calculating the parameters, and updating the settings accordingly. For verification purposes, Matlab SimPower-Systems® was utilized, employing a straightforward radial topology. However, the authors acknowledge the need for more complex scenarios to be included in future analyses and investigations. In a subsequent work [92], a modification of the methodology presented in [70]. was proposed. This modification involves the inclusion of a central controller agent responsible for identifying the post-fault network configuration through communication with other agents. The remaining agents determine the network configuration by observing pre-fault conditions. Kiani et al. [93] addressed the time-delay issue commonly encountered in multi-agent structures by incorporating a multi-thread process that employed different fault-clearing algorithms for various scenarios. This approach effectively resolved the problem of time delay and eliminated the reliance on a central controller during fault conditions. The authors also identified several areas for future research, including the coordination of fuse-relay devices, development of a more reliable scheme that accounts for multiple faults, proposal of an extended approach for loop networks, and presentation of a similar approach to HIF. In more recent research, such as [94,95], additional hardware-in-the-loop (HIL) and embedded testbeds have been implemented. The authors in [94] achieved decentralized very fast autonomous operation even with loss of communication, although no fault classification, fault location or multi microgrid capabilities were included. The results in [95] presented uncertainty regarding what would occur during a communication loss. In addition, fault classification or location were not addressed. High accuracy with fault location capability can be found in [53]; though including hardware acceleration for training the proposed deep neural network. The authors in [96] described a scheme combining fault detection and power restoration. Table 3 provides an overview of the adaptive MAS-based approaches reviewed in this manuscript. Fault location techniques are typically classified into four main categories: (i) traveling wave-based methods; (ii) soft computing techniques, ANN and ML approaches; (iii) transient fault analysis of high-frequency signals; and (iv) phasor-based algorithms that rely on fundamental frequency signals. The reliance of traditional fault location algorithms on fault classification data diminishes the reliability of fault location schemes, making these

Table 3
Summary of recent MAS adaptive protection approaches.

Ref.	Year	Architecture	Scenario	Communications Infrastructure	Comms. protocol [2]	Fault types	Fault trip time (ms)[1]	Power simulation	MAS framework	Agent designations	Advantages	Drawbacks
[96]	2024	DT	Is	(–)	(–)	LG, LLL	(–)	MATLAB/Simulink	(–)	OA, FLA, ReA,	Combined fault detection and power restoration	– No fault classification, location or multi microgrid tested – Uncertainty in communication loss
[53]	2024	DT	GT, Is	None	None	LG, LL LLL	(–)	DIgSILENT, MATLAB	Python, Tensorflow	PA	High accuracy in fault detection and fault location	– Computation intensive neural network training required
[94]	2023	DT	GT, Is	(–)	(–)	LG, LLG, LLL	7.02–11.18	PSCAD/EMTD	Simulink, C code	SeA, SA, CBA	Autonomous, HIL tested and fast response	– No fault classification, location or multi microgrid tested
[95]	2023	DT	GT, Is	Ethernet/TCP-IP	IEC 60870-5–104 /Modbus	LLL	30–1107	ETAP, Hypersim software, OPAL-RT	JADE	MA, PA	HIL tested	– Uncertainty in communication loss – No fault classification or location
[90]	2022	DT	GT, Is	(–)	(–)	LG, LL	153	MATLAB	JADE	RA	Low relative complexity, selective and sensitive	– Uncertainty during cyberattacks
[12]	2022	DT	GT, Is	Ethernet	IEC 61850		150–800	MATLAB/Simulink		RA, MA, CBA, CMA	Reduced communication delay due to offline calculations	– Only offline calculations implemented. Uncertain performance in complex topology changes
[16]	2021	DT	GT, Is	Ethernet	IEC 61850 GOOSE SMV	(–)	120	ETAP	(–)	DGA	Preservation of coordination during settings update	– No topology change detection
[115]	2021	Hy	GT, Is	(–)	FIPA	LG	70–1437	PSCAD/EMTD	JADE	RA CoA, CA	Backup protection during communication loss	– Uncertainty during cyber attack and increased cost due to backup protection
[88]	2021	DT	GT, Is	VLAN Ethernet	IEC 61850 GOOSE SV	LLLG LLG	100	ETAP, MATLAB	(–)	ORA	Very capable in various topology changes	– Needs testing large microgrid scenarios
[116]	2019	DT	GT, Is	(–)	(–)	L, LL LLL	482–636	MATLAB	(–)	RA, DGA PCCA, VTA	Capable in topology changes	– Uncertainty in communication loss
[87]	2019	DT	GT, Is	(–)	(–)	LL, LG LLG	(–)	MATLAB/Simulink	AnyLogic	MA, PA LA, VA	Combined benefits of MAS and machine learning intelligence	– Requires real-time verification and computation intensive
[86]	2019	CT	(–)	(–)	TCP/IP	LG LLG LLLG	(–)	MATLAB/Simulink	JADE	CBA	Reliability via Q-learning algorithm and information protection via blockchain	– Compute intensive and Communication dependent
[85]	2019	DT	(–)	Ethernet	IEC 61850 GOOSE SMV	LG LLG LLL	800	ETAP	(–)	RA, DGA	Reliability and reduced network traffic	– Network data collection and analysis is time consuming – Vulnerable during relay settings update
[82]	2019	DT	(–)	(–)	(–)	(–)	(–)	MATLAB	NetLogo	FA, SA ZA, DGA	Fault restoration	– Higher cost and redundant
[117]	2018	DT	GT, Is	(–)	(–)	LG LLG	(–)	RSCAD	JADE	MA PCCA RA, CBA	Effective fault detection	– Communication loss uncertainty – DG penetration not considered – No real-time testing
[91]	2018	DT	GT, Is	(–)	(–)	LLL	50	MATLAB/ SimPowerSystems	(–)	RA PCCA DGA	Versatile configuration and coordination	– Heavy on communication and memory system
[92]	2018	Hy	GT, Is	(–)	(–)	LLL	50	MATLAB/ SimPowerSystems	(–)	RA PCCA, DGA MGCA	Versatile reconfiguration and coordination	– Heavy on communication and memory system

(*continued on next page*)

Table 3 (*continued*)

Ref.	Year	Architecture	Scenario	Communications Infrastructure	Comms. protocol [2]	Fault types	Fault trip time (ms)[1]	Power simulation	MAS framework	Agent designations	Advantages	Drawbacks
[98]	2013	Hy	GT, Is	Ethernet	IEC 61850	LG	140–361	PSCAD/EMTDC	JADE	RA, CoA, CA	Effective for static small-scale MG	Not best for large dynamic MG

Legend approach:
Challenge Focus: Coordination (COR), Selectivity (SEL)
Architecture: Distributed (DT), Centralized (CT), Hybrid (Hy)
Topology: Grid-tied (GT), Islanded (Is)
Communications Infrastructure
Fault types: Single-phase (L), single-phase (L), Three-phase (LLL), Two-phase (LL), to ground (G)
Agent types: Distributed generator agent (DGA), relay agent (RA), coordinator agent (CoA), configurator agent (CA), overcurrent relay agent (ORA), Point of common coupling agent (PCCA), Voltage transformer agent (VTA), measurement agent (MA), protection agent (PA), local agent (LA), circuit breaker agent (CBA), Feeder Agents (FA), Switch Agents (SA), Zone Agents (ZA), MicroGrid Central Controller Agent (MGCA), Communications Agent (CMA), Observer Agent OA, Fault LimiterAgent (FLA), Restoring Agent (ReA), Protection Agent (PA), SeA
Not specified (–)

[1] In many of the papers, the trip time provided depended on the relay in the bus structure and on whether it was in an online or offline operation.
[2] Different authors use the terms SV (Sampled values) and SMV (Sampled measure values) as referring to the same protocol.

algorithms heavily dependent on accurate fault classification. More recently, Wakar et al. [97] introduced a straightforward and innovative microgrid protection method that combines superimposed components, the Wigner Distribution Function (WDF), and an alienation index-based approach.

Regarding the communication infrastructure, it currently plays a crucial role in MAS-based protection approaches. It is imperative to use a standardized protocol specifically designed for power distribution purposes. Habib et al. [39] reviewed in detail the communication requirements for protection algorithms under the IEC 61850 international standard. Developed by the International Electrotechnical Commission (IEC) Technical Committee Number 57, Working Group 10, in collaboration with the IEEE, the IEC 61850 standard focuses on Ethernet (IEEE 802.3)-based communication in power system substations. This fact is also evident in most reviewed contributions, such as [16,98]. Indeed, its scope has been expanded to encompass tele-protection (IEC 61850–90-1). Recent literature has also highlighted the relevance of the IEC 61850 standard in MG applications. The IEC 61850 standard imposes strict limitations on fault-related signal communication. For instance, it enforces a 4 ms time constraint on the sampled measured values (SMV) and generic object-oriented substation event (GOOSE) messages. Prior power system applications relied predominantly on the power-line communication (PLC), as highlighted in [13], before the widespread adoption of wireless technologies. However, wireless protocols, such as the ZigBee, WLAN, cellular communication, and WiMAX, which are currently considered in power system applications [36], are not suitable for the MG protection due to their higher latency in comparison to the maximum 4 ms latency requirement for the MG protection information [99,100]. Rath et al. [101] recently affirmed that the majority of these communication protocols lack end-to-end encryption, leaving them exposed to potential cyberattacks and being necessary further contributions to overcome such potential vulnerabilities.

As was previously discussed, modern power systems have been transformed into multidimensional heterogeneous cyber-physical systems [102]. Based on this concept, the control structure is divided into two layers: the cyber layer and physical layer. With this evolution, the threat of cyberattacks can have a significant impact on the power systems by manipulating critical infrastructure information [103]. Indeed, the communication architecture must provide end-to-end security, protect the MGs from potential cyberattacks, maintain power control functions with low-latency communication, and ensure scalability for future growth [104]. In [105], a response hierarchical control strategy is proposed and evaluated including communication data disturbances. To set up the experimental power system platform, including all the components, multiple software solutions, such as Matlab/Simulink, PSCAD, DigSILENT, and ETAP were proposed [10,18]. Advancements in digital processing technology have also led to the increased use of HIL tests for power system real-time simulations [63,106]. Nevertheless, the use of the HIL platforms was identified in earlier studies [107]. Development frameworks for applying agent systems are based on various software platforms. NetLogo was used in the literature review [82]. Another framework identified in various studies is the JADE [98]. Lastly, AnyLogic [87] was identified as an applicable agent simulation environment. Although various agent development and simulation tools have been identified, it is evident that JADE is the most widely used tool in recent building MAS contributions. In earlier works, the hardware implementation of an MAS-based fault location solution was also reported [108]. From the specific literature, it can be affirmed that HIL is increasingly gaining traction as an effective test bed for evaluating smart grids and MG development [10,63,109].

Potential communication link failures or interruptions can render these schemes ineffective. As a result, certain non-pilot methods have been explored in the technical literature, but they often fail to perform adequately in the presence of inverter-based resources due to the distinct fault characteristics of them compared to traditional synchronous generators. To overcome these potential ineffective schemes,

recent contributions proposed an accelerated distance protection scheme for the lines connected to inverter-based resources [110]. In [111], a distributed control algorithm was proposed for secondary voltage and frequency restoration in an autonomous inverter-based MG, accounting for simultaneous state-dependent noise and potential communication time delays. In [112], distributed economic operation of islanded AC MGs under uncertain communication links is studied and evaluated. In [113], a hybrid MAS design is integrated with a peer-to-peer control system, where agents maintain a record of both available and unavailable communication links, as well as nearby isolated regions. In addition, as IoT applications become increasingly prevalent and digital relays are integrated into IoT systems, protection systems can communicate with adjacent relays through wired and wireless communication technologies. This connectivity will enable the protection system to adapt based on surrounding or abnormal conditions [114].

5. Discussion

The reviewed adaptive protection approaches relied on a centralized architecture and heavy use of fault detection based on overcurrent. In contrast, other detection methods, such as distance and differential relays, were much less employed in the reviewed adaptive protection literature. Most research contributions presented extensive use of a communications infrastructure. The abovementioned was evident in proposals with a centralized and hybrid architecture. Regarding such works that used a distributed architecture, around 50 % of them utilized communications in their scheme. The rest of the contributions did not use communications, being this group the one with the most recent solutions. Consequently, the abovementioned indicates a trend toward reducing the need for communications. This point can be graphically seen in Fig. 4 that shows a classification of the reviewed literature based on architecture. The most widely utilized communications protocol for MG adaptive protection, especially MAS-based approaches, was IEC 61850. It is not unexpected since it was specially developed to supervise and control power distribution systems. Ethernet is a well-known and established technology, which simplifies its application. However, this simplicity might also give way to increased cybersecurity threats. Note that no other types of networks could be identified in the reviewed literature.

The consideration of other communication network technologies applied to MAS-based systems, would be a field of interest by the scientific community. Relevant contributions discussed in Section 3 suggested future research work in MAS-based MG protection to address scalability and communication failure issues. Some of such works mentioned the need to tackle real-time performance, real-world implementation, and cybersecurity threats. It seems these challenges remain active in the current research landscape for MAS-Based MG protection. MAS-based protection schemes possess a predominantly distributed architecture instead of a centralized solution, though some recent proposals apply a hybrid approach, as can be seen in Fig. 5. Further use of hybrid models in the MAS architecture might provide a more optimized solution while reducing the possibility of a single point of failure. Tazi et al. [118] affirmed that most studies assume a homogeneous MAS architecture to simplify handling system complexity, while the optimization and behavior of heterogeneous MAS remains an intriguing research opportunity.

Although various agent development and simulation tools have been identified, it was evident that JADE was the most widely used framework for MAS solutions in recent contributions. JADE is an open development framework built on the well-known JAVA programming language. Its use is expected to increase further in MAS-based MG protection development. The agents were primarily found implemented in IEDs in the form of RA, DGA, and PCC, though other device types with IED capabilities were also applied, as can be seen in Fig. 6. Note that such contributions with RAs had the widest variety of faults addressed due to the wide availability and well-understood operation. Additionally, while being an advantageous asset in modern smart grid and MG stability applications, ESSs were not seen in the various agent types in the specific literature. This figure also shows that single and three-phase faults were considered in most reviewed works. Ground faults were considered as well, but to a slightly lesser extent. In these contributions, there was no mention of addressing the issue of high impedance faults. The above-mentioned should be included in future works for a more complete solution to MG protection.

Most of the power systems included in contributions of Table 3 (see Section 4) were based on software simulation, such as with MATLAB or ETAP, but implementations of real-world practical examples with real-time execution were scarce. It would not be a simple task, since the IEDs associated with the agents would be required to have remarkable

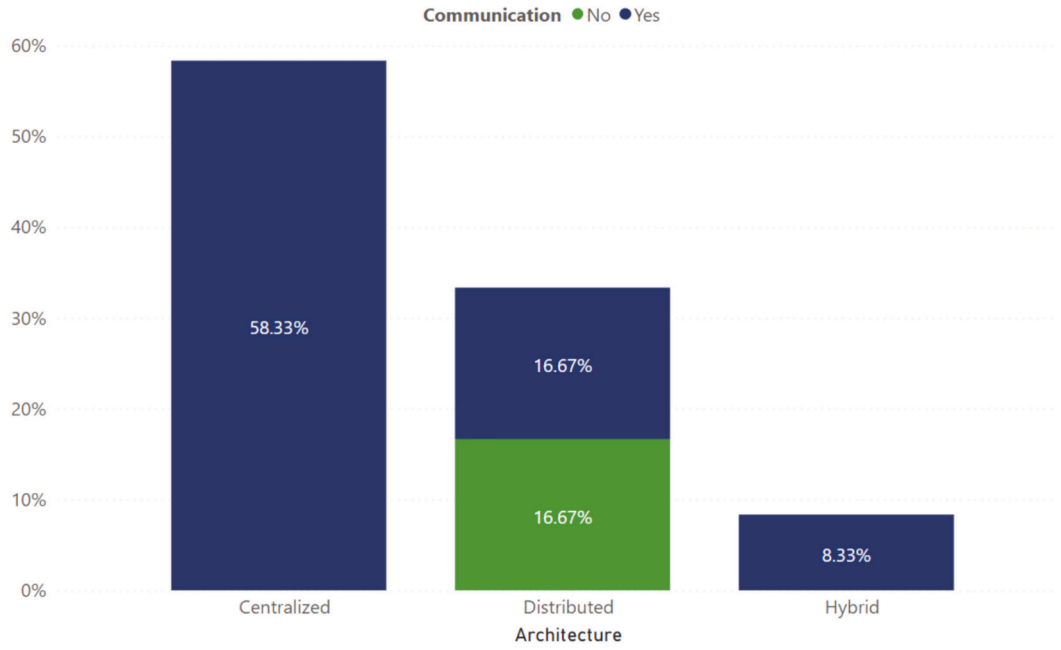


Fig. 4. Architecture vs. communication.

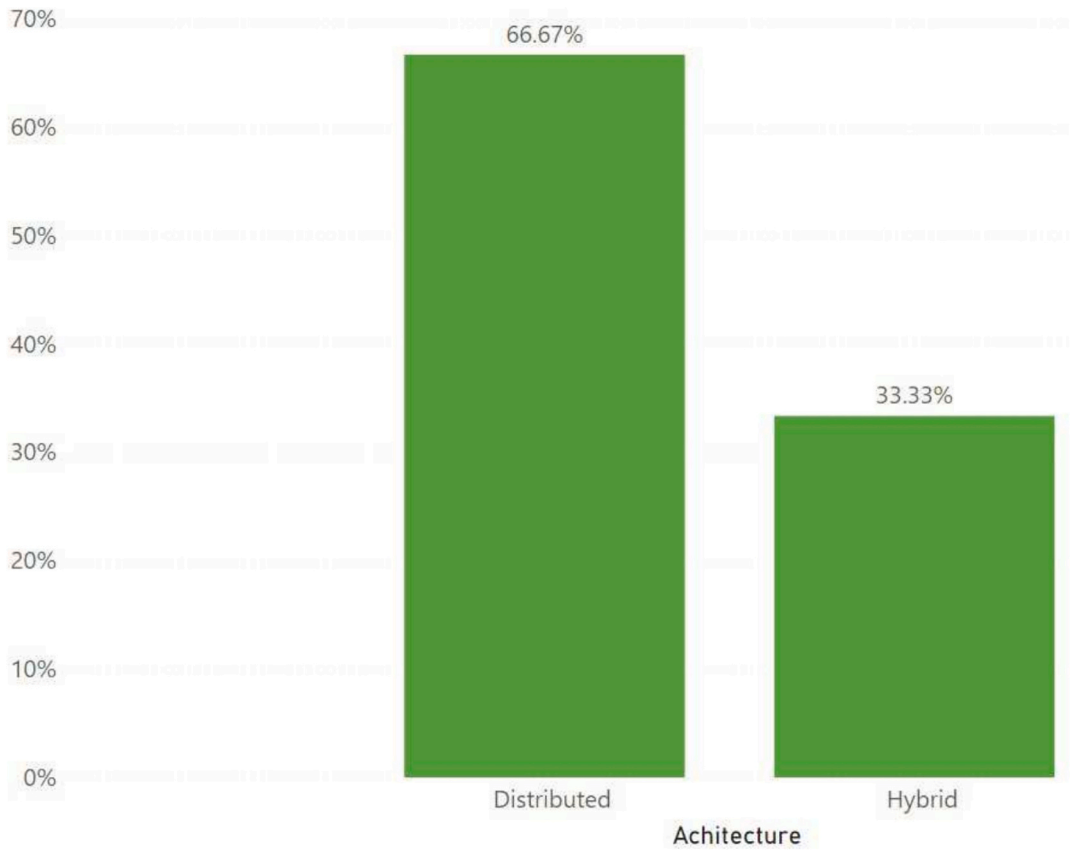


Fig. 5. MAS-based protection schemes.

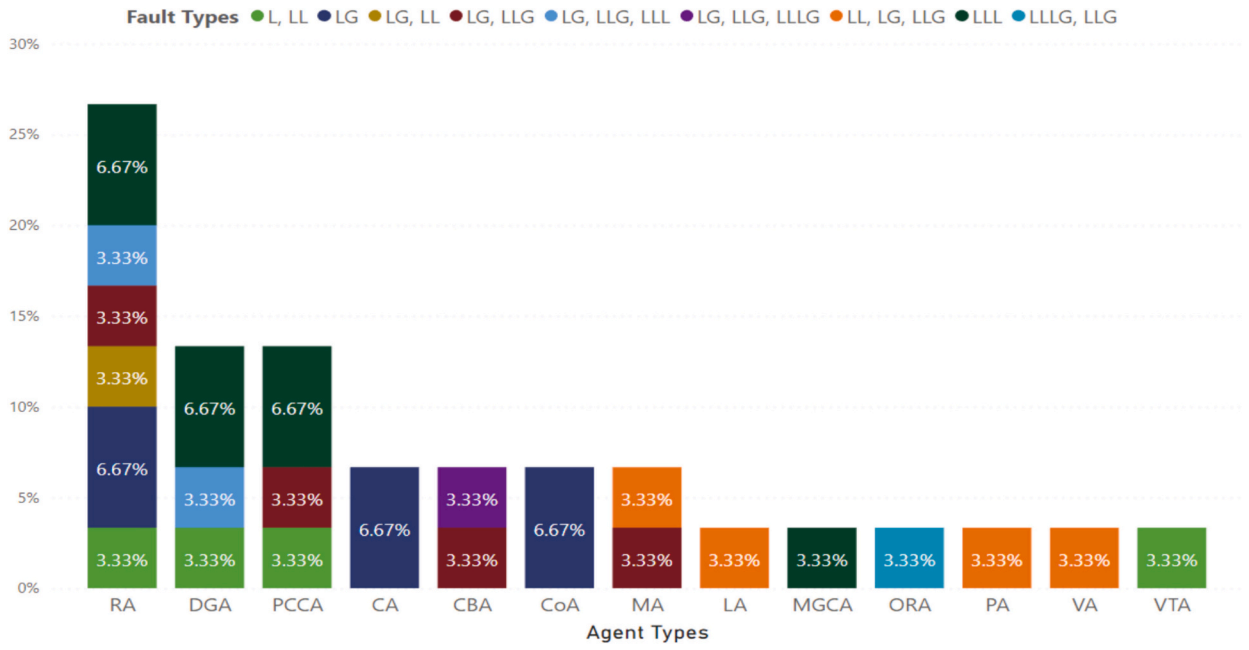


Fig. 6. MAS fault type distribution.
Agent types: Relay agent (RA), distributed generator agent (DGA), point of common coupling agent (PCCA), configurator agent (CA), circuit breaker agent (CBA), coordinator agent (CoA), measurement agents (MA), local agent (LA), MicroGrid Central Controller Agent (MGCA), overcurrent relay agent (ORA), protection agent (PA), voltage agent (VA), Voltage transformer agent (VTA).

computational capabilities to execute the corresponding algorithms. Consequently, identifying real-world devices that are part of the MG and with the local processing capacity would be increasingly complex and costly. However, an opportunity is recognized in validating novel MAS-based adaptive protection schemes using advanced HIL platforms (with increased computational capability and scalability).

6. Future work

To further develop the application of MAS-based adaptive protection systems for MGs, additional studies must be conducted to overcome the current limitations. Most of the reviewed contributions provided experimental tests based on software simulation environments with non-real-time performance features. MGs are complex cyber-physical systems, for which near-real-time operation, if not completely real time, is critically important. Consequently, this capability of all the agent types and abstraction layers of the MAS-based approach must be considered in future research. MAS solutions can operate in centralized, distributed, or hybrid structures. This characteristic of MAS distributed operation is a key factor to point out the interest around them. Therefore, autonomy at different levels during communication failures presents a considerable interest in the reviewed literature. This fact most likely does not change because autonomy at different levels improves the reliability and resiliency of the protection system. To maximize their capabilities, they rely heavily on robust communication architectures. Therefore, further studies should be conducted to reduce the burden of the MAS-based strategies on the communication systems, but at the same time, increasing the number of different types of networks used to deploy MAS-based adaptive protection. Future applications may also enable real-time control of individual relays. Additionally, cybersecurity is of paramount importance and concern. Consequently, it is a topic that requires more attention. The cybersecurity application in the microgrid MAS system should account for its ability to detect and defend against attacks and its effects on the protection system's performance. The resilience against cyber-attacks, for example, denial-of-service (DoS) attacks, can be also found in recent contributions [119].

Note that there is an opportunity for further research in increasing the number of faults that can be addressed with agents other than RA, DGA, and PCC. Moreover, it would be interesting to consider discussing and addressing HIF in MAS-based adaptive protection solutions. Another subject that should be further considered is the potential role of ESS as agents in MAS-based adaptive protection systems, as there is scarce evidence of its application in these schemes. Few recent contributions, such as in [44], considered ESS agents in their MAS-based MG protection approach. However, further works would be necessary in this field. The MAS-based adaptive protection scheme should be able to apply under different combinations of fault detection methods, apart from overcurrent. Additional contributions focused on improving the computational performance of MAS algorithm implementations should also be of interest for future works. These future proposals can facilitate the integration of lower-powered embedded processors, commonly found in the devices deployed in MGs and the rest of the main power grid, promoting better integration of MAS algorithms into the equipment layer. Additionally, decreasing the cost of MAS-based adaptive protection strategies should be a favorable issue.

7. Conclusion

MGs are a growing research topic and field of interest for the scientific community. Their successful implementation would support the increased application of the DGs, especially those generations based on renewable energy resources. With the penetration of DG, especially IBDG, it is important that protection strategies adjust themselves to the dynamic interaction and behavior of MG systems. A description of the background to the application of MGs, specific protection challenges, and solutions for AC MGs, particularly focusing on adaptive protection based on MASs, is discussed in this review. Adaptive protection schemes are described as effective solutions to the challenges presented in this study. Among these schemes, and in line with the various approaches mentioned in the literature, the one that relies on the application of MASs stands out. MAS-based adaptive protection approaches can address multiple types of faults, either single- or multiple-events, and multiple types of topology scenarios. This method also appropriately addresses the miscoordination among the protection relays, though more research would be required for MAS-based solutions. They are also highly autonomous, tolerant of single-point failures, and robust. Nevertheless, some studies have found that high-impedance faults have not been satisfactorily addressed.

Communication infrastructure is critical for optimal deployment of modern protection schemes. The most used protocol is the IEC 61850. MAS-based adaptive protection is currently expensive, primarily because of the communication and multidevice agent architecture. However, this fact might be of minor concern, as an increasing number of devices are currently available with smart communication and computational capabilities as standard features. New advances in device development and manufacturing reduce equipment costs and encourage key decision-makers to assign resources to develop this research topic further.

Recent literature has identified the scalability of the MASs as an issue that is clearly limited. This is because of the experimental observations included in such contributions, in which it was determined that increasing the number of agents would increase the computational requirements, and thus, raise the cost accordingly. The authors identified a lack of recent primary research on the MAS-based adaptive protection, even when considering the use of ESSs as active agents in the protection scheme. In specific situations, MAS systems can produce imbalances owing to abrupt disconnections from the main grid, which can be quickly detected by the corresponding protection systems. Consequently, AC microgrid protection based on multi-agent systems requires further research regarding scalability, real-time performance, real-world testing, cybersecurity threat mitigation, and cost reduction. Moreover, there is a need for more research on fault detection methods other than overcurrent, as well as increasing the number of types of combined agents and the types of communication networks applied in MAS-based microgrid adaptive protection systems.

CRediT authorship contribution statement

Armando J. Taveras-Cruz: Resources, Project administration, Conceptualization. **Deyslen Mariano-Hernández:** Writing – original draft, Methodology, Formal analysis. **Elvin Jiménez-Matos:** Writing – original draft, Software, Resources, Formal analysis. **Miguel Aybar-Mejia:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Patricio A. Mendoza-Araya:** Writing – review & editing, Supervision, Conceptualization. **Angel Molina-García:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

[1] Dagar A, Gupta P, Niranjan V. Microgrid protection: A comprehensive review. Renew Sust Energ Rev 2021;149:111401. https://doi.org/10.1016/j.rser.2021.111401.

[2] CIGRÉ W group C 22. Microgrids 1: Engineering. Economics, & Experience. In: CIGRE Session Technical Brochure; 2015. p. 144.

[3] Brearley BJ, Prabu RR. A review on issues and approaches for microgrid protection. Renew Sust Energ Rev 2017;67:988–97. https://doi.org/10.1016/J.RSER.2016.09.047.

[4] Altaf MW, Arif MT, Islam SN, Haque MdE. Microgrid protection challenges and mitigation approaches–A comprehensive review. IEEE Access 2022;10:38895–922. https://doi.org/10.1109/ACCESS.2022.3165011.

[5] Patnaik B, Mishra M, Bansal RC, Jena RK. AC microgrid protection – A review: current and future prospective. Appl Energy 2020;271:115210. https://doi.org/10.1016/j.apenergy.2020.115210.

[6] Manditereza PT, Bansal R. Renewable distributed generation: the hidden challenges – A review from the protection perspective. Renew Sust Energ Rev 2016;58:1457–65. https://doi.org/10.1016/j.rser.2015.12.276.

[7] Li X, Li Z, Guo L, Zhu J, Wang Y, Wang C. Enhanced dynamic stability control for low-inertia hybrid AC/DC microgrid with distributed energy storage systems. IEEE Access 2019;7:91234–42. https://doi.org/10.1109/ACCESS.2019.2926814.

[8] Gadanayak DA. Protection algorithms of microgrids with inverter interfaced distributed generation units—A review. Electr Power Syst Res 2021;192:106986. https://doi.org/10.1016/j.epsr.2020.106986.

[9] Abasi M, Rohani A, Hatami F, Joorabian M, Gharehpetian GB. Fault location determination in three-terminal transmission lines connected to industrial microgrids without requiring fault classification data and independent of line parameters. Int J Electr Power Energy Syst 2021;131:107044. https://doi.org/10.1016/j.ijepes.2021.107044.

[10] Shobole AA, Wadi M. Multiagent systems application for the smart grid protection. Renew Sust Energ Rev 2021:149. https://doi.org/10.1016/j.rser.2021.111352.

[11] Kaur G, Prakash A, Rao KU. A critical review of microgrid adaptive protection techniques with distributed generation. Renew Energy Focus 2021;39:99–109. https://doi.org/10.1016/J.REF.2021.07.005.

[12] Aazami R, Esmaeilbeigi S, Valizadeh M, Javadi MS. Novel intelligent multi-agents system for hybrid adaptive protection of micro-grid. Sustain Energy, Grids Networks 2022;30:100682. https://doi.org/10.1016/j.segan.2022.100682.

[13] Senarathna TSS, Udayanga Hemapala KTM. Review of adaptive protection methods for microgrids. AIMS Energy 2019;7:557–78. https://doi.org/10.3934/energy.2019.5.557.

[14] Damanjani A, Abardeh MH, Azarfar A, Hojjat M. A statistics-based review of microgrid protection with a concentration on adaptive protection. Int Rev Appl Sci Eng 2021;12:312–23. https://doi.org/10.1556/1848.2021.00306.

[15] Khalid H, Shobole A. Existing developments in adaptive smart grid protection: A review. Electr Power Syst Res 2021:191. https://doi.org/10.1016/j.epsr.2020.106901.

[16] Abbaspour E, Fani B, Sadeghkhani I, Alhelou HH. Multi-Agent System-based hierarchical protection Scheme for distribution networks with high penetration of electronically-coupled DGs. IEEE Access 2021;9:102998–3018. https://doi.org/10.1109/ACCESS.2021.3098387.

[17] Hsiao J-H, Chen K-C. Communication methodology to control a distributed multi-agent system. In: ICC 2019-2019 IEEE International Conference on Communications (ICC). IEEE; 2019. p. 1–6. https://doi.org/10.1109/ICC.2019.8761869.

[18] Barra PHA, Coury DV, Fernandes RAS. A survey on adaptive protection of microgrids and distribution systems with distributed generators. Renew Sust Energ Rev 2020;118:109524. https://doi.org/10.1016/j.rser.2019.109524.

[19] Altaf MW, Arif MT, Islam SN, Haque MdE. Microgrid protection challenges and mitigation approaches–A comprehensive review. IEEE Access 2022;10:38895–922. https://doi.org/10.1109/ACCESS.2022.3165011.

[20] Patnaik B, Mishra M, Bansal RC, Jena RK. AC microgrid protection – A review: current and future prospective. Appl Energy 2020;271:115210. https://doi.org/10.1016/j.apenergy.2020.115210.

[21] Uzair M, Li L, Eskandari M, Hossain J, Zhu JG. Challenges, advances and future trends in AC microgrid protection: with a focus on intelligent learning methods. Renew Sust Energ Rev 2023;178:113228. https://doi.org/10.1016/j.rser.2023.113228.

[22] Shuai Z, Sun Y, Shen ZJ, Tian W, Tu C, Li Y, et al. Microgrid stability: classification and a review. Renew Sust Energ Rev 2016;58:167–79. https://doi.org/10.1016/j.rser.2015.12.201.

[23] Shrivastava S, Subudhi B. Comprehensive review on hierarchical control of cyber-physical microgrid system. IET Gener Transm Distrib 2020;14:6397–416. https://doi.org/10.1049/iet-gtd.2020.0971.

[24] Chandak S, Rout PK. The implementation framework of a microgrid: A review. Int J Energy Res 2021;45:3523–47. https://doi.org/10.1002/er.6064.

[25] Brearley BJ, Prabu RR. A review on issues and approaches for microgrid protection. Renew Sust Energ Rev 2017;67:988–97. https://doi.org/10.1016/J.RSER.2016.09.047.

[26] Vegunta SC, Higginson MJ, Kenarangui YE, Li GT, Zabel DW, Tasdighi M, et al. AC microgrid protection system design challenges—A practical experience. Energies (Basel) 2021:14. https://doi.org/10.3390/en14072016.

[27] Kaur G, Prakash A, Rao KU. A critical review of microgrid adaptive protection techniques with distributed generation. Renew Energy Focus 2021;39:99–109. https://doi.org/10.1016/J.REF.2021.07.005.

[28] Beheshtaein S, Cuzner R, Savaghebi M, Guerrero JM. Review on microgrids protection. IET Gener Transm Distrib 2019;13:743–59. https://doi.org/10.1049/iet-gtd.2018.5212.

[29] Gadanayak DA, Mallick RK. Microgrid differential protection scheme using downsampling empirical mode decomposition and Teager energy operator. Electr Power Syst Res 2019;173:173–82. https://doi.org/10.1016/j.epsr.2019.04.022.

[30] Fathima F, Premalatha L. Protection strategies for AC and DC microgrid – A review of protection methods adopted in recent decade. IETE J Res 2021:1–17. https://doi.org/10.1080/03772063.2021.1990140.

[31] Aboelezz AM, Sedhom BE, El-Saadawi MM. Intelligent distance relay based on IEC 61850 for DC zonal shipboard microgrid protection. In: 2022 second international conference on power, control and computing technologies (ICPC2T). IEEE; 2022. p. 1–5. https://doi.org/10.1109/ICPC2T53885.2022.9776712.

[32] Tsimtsios George N, Nikolaidis Vassilis C, AM; K.. A pilot-based distance protection scheme for meshed distribution systems with distributed generation. Int J Electr Power Energy Syst 2019;105:454–69. https://doi.org/10.1016/j.ijepes.2018.08.022.

[33] Kou G, Jordan J, Cockerham B, Patterson RW, VanSant P. Negative-sequence current injection of transmission solar farms. IEEE Trans Power Deliv 2020:1. https://doi.org/10.1109/TPWRD.2020.3014783.

[34] Bera PK, Pani SR, Isik C, Bansal RC. A hybrid intelligent system for protection of transmission lines connected to PV farms based on linear trends. Electr Power Syst Res 2024;237:110991. https://doi.org/10.1016/j.epsr.2024.110991.

[35] Alasali F, Saad SM, Saidi AS, Itradat A, Holderbaum W, El-Naily N, et al. Powering up microgrids: A comprehensive review of innovative and intelligent protection approaches for enhanced reliability. Energy Rep 2023;10:1899–924. https://doi.org/10.1016/j.egyr.2023.08.068.

[36] Chandra A, Singh GK, Pant V. Protection of AC microgrid integrated with renewable energy sources – A research review and future trends. Electr Power Syst Res 2021;193:107036. https://doi.org/10.1016/J.EPSR.2021.107036.

[37] Brahmendra Kumar GV, Palanisamy K. Review of energy storage system for microgrid. Microgrid Technologies, Wiley; 2021. p. 57–90. https://doi.org/10.1002/9781119710905.ch3.

[38] Joshua AM, Vittal KP. Protection schemes for a battery energy storage system based microgrid. Electr Power Syst Res 2022;204:107701. https://doi.org/10.1016/j.epsr.2021.107701.

[39] Habib HF, Lashway CR, Mohammed OA. A review of communication failure impacts on adaptive microgrid protection schemes and the use of energy storage as a contingency. IEEE Trans Ind Appl 2018;54:1194–207. https://doi.org/10.1109/TIA.2017.2776858.

[40] Chandra A, Singh GK, Pant V. Protection techniques for DC microgrid- A review. Electr Power Syst Res 2020;187:106439. https://doi.org/10.1016/j.epsr.2020.106439.

[41] Jayawarna N, Barnes M. Central storage unit response requirement in 'good citizen' Microgrid. In: 2009 13th European Conference on Power Electronics and Applications; 2009. p. 1–10.

[42] Habib HF, Esfahani MM, Mohammed O. Improvement of protection scheme for microgrids using lithium- ion battery during islanding. In: 2018 IEEE industry applications society annual meeting (IAS). IEEE; 2018. p. 1–8. https://doi.org/10.1109/IAS.2018.8544559.

[43] Habib HF, Mohamed AAS, El Hariri M, Mohammed OA. Utilizing supercapacitors for resiliency enhancements and adaptive microgrid protection against communication failures. Electr Power Syst Res 2017;145:223–33. https://doi.org/10.1016/j.epsr.2016.12.027.

[44] Fawzy N, Habib HF, Mohammed O, Brahma S. Protection of microgrids with distributed generation based on multiagent system. In: 2020 IEEE international conference on environment and electrical engineering and 2020 IEEE industrial and commercial power systems Europe (EEEIC / I&CPS Europe). IEEE; 2020. p. 1–5. https://doi.org/10.1109/EEEIC/ICPSEurope49358.2020.9160827.

[45] Hatziargyriou N. Microgrids: architectures and control. John Wiley & Sons; 2014.

[46] Antonova G, Nardi M, Scott A, Pesin M. Distributed generation and its impact on power grids and microgrids protection. In: 2012 65th annual conference for protective relay engineers. IEEE; 2012. p. 152–61. https://doi.org/10.1109/CPRE.2012.6201229.

[47] Ilyushin P, Suslov K. Operation of automatic transfer switches in the networks with distributed generation. In: 2019 IEEE Milan PowerTech. IEEE; 2019. p. 1–6. https://doi.org/10.1109/PTC.2019.8810450.

[48] Phadke AG, Thorp JS. Computer relaying for power systems. 2nd ed. Hoboken, NJ: Wiley; 2009. https://doi.org/10.1002/9780470749722.

[49] Dy Liacco T. The adaptive reliability control System. IEEE Trans Power Apparatus Syst 1967;PAS-86:517–31. https://doi.org/10.1109/TPAS.1967.291728.

[50] Shih MY, Conde A, Leonowicz Z, Martirano L. An adaptive overcurrent coordination Scheme to improve relay sensitivity and overcome drawbacks due to Distributed generation in smart grids. IEEE Trans Ind Appl 2017;53:5217–28. https://doi.org/10.1109/TIA.2017.2717880.

[51] Shah R, Goli P, Shireen W. Adaptive protection scheme for a microgrid with high levels of renewable energy generation. In: 2018 Clemson University power systems conference (PSC). IEEE; 2018. p. 1–7. https://doi.org/10.1109/PSC.2018.8664068.

[52] Bakkar M, Bogarra S, Córcoles F, Aboelhassan A, Wang S, Iglesias J. Artificial intelligence-based protection for smart grids. Energies (Basel) 2022:15. https://doi.org/10.3390/en15134933.

[53] Najar A, Kazemi Karegar H, Esmaeilbeigi S. Multi-agent protection scheme for microgrid using deep learning. IET Renew Power Gener 2024;18:663–78. https://doi.org/10.1049/rpg2.12929.

[54] Samal S, Samantaray SR, Manikandan MS. A DNN based intelligent protective relaying Scheme for microgrids. In: 2019 8th international conference on power systems (ICPS). IEEE; 2019. p. 1–6. https://doi.org/10.1109/ICPS48983.2019.9067600.

[55] Peter N, Gupta P, Goel N. Fault detection and identification of fault location in hybrid microgrid using artificial neural network. In: 2023 10th international conference on signal processing and integrated networks (SPIN). IEEE; 2023. p. 686–91. https://doi.org/10.1109/SPIN57001.2023.10116186.

[56] Singh M, Basak P. Adaptive protection methodology in microgrid for fault location and nature detection using q 0 components of fault current. IET Gener Transm Distrib 2019;13:760–9. https://doi.org/10.1049/iet-gtd.2018.6347.

[57] Gu J, Liu C, Wang J, Yang M. Using IEC 61850 GOOSE messages in microgrid protection. Int Trans Electr Energy Syst 2019;29. https://doi.org/10.1002/2050-7038.12122. NA-NA.

[58] Dorosti P, Moazzami M, Fani B, Siano P. An adaptive protection coordination scheme for microgrids with optimum PV resources. J Clean Prod 2022:340. https://doi.org/10.1016/j.jclepro.2022.130723.

[59] Kai Lin, Zheng-Hua Tan, Chengxi Liu, Guerrero Josep M, CHS Vasquez Juan. Adaptive protection combined with machine learning for microgrids. IET Gener Transm Distrib 2019;13:770–9. https://doi.org/10.1049/iet-gtd.2018.6230.

[60] Brearley BJ, Raja Prabu R, Regin Bose K, Sankaranarayanan V. Adaptive relay coordination scheme for radial microgrid. Int J Ambient Energy 2022;43:2180–93. https://doi.org/10.1080/01430750.2020.1722226.

[61] Sati TE, Azzouz MA. An adaptive virtual impedance fault current limiter for optimal protection coordination of islanded microgrids. IET Renew Power Gener 2022;16:1719–32. https://doi.org/10.1049/rpg2.12474.

[62] El-Naily N, Saad SM, Elhaffar A, Zarour E, Alasali F. Innovative adaptive protection approach to maximize the security and performance of phase/earth overcurrent relay for microgrid considering earth fault scenarios. Electr Power Syst Res 2022:206. https://doi.org/10.1016/j.epsr.2022.107844.

[63] Naveen P, Jena P. Adaptive protection scheme for microgrid with multiple point of common couplings. IEEE Syst J 2021;15:5618–29. https://doi.org/10.1109/JSYST.2020.3039881.

[64] Chae W, Lee J-H, Kim W-H, Hwang S, Kim J-O, Kim J-E, et al. Adaptive protection coordination method design of remote microgrid for three-phase short circuit fault. 2021. https://doi.org/10.3390/en14227754.

[65] Saldarriaga-Zuluaga SD, López-Lezama JM, Muñoz-Galeano N. Adaptive protection coordination scheme in microgrids using directional over-current relays with non-standard characteristics. Heliyon 2021;7:e06665. https://doi.org/10.1016/J.HELIYON.2021.E06665.

[66] Ghadiri SME, Mazlumi K. Adaptive protection scheme for microgrids based on SOM clustering technique. Appl Soft Comput J 2020;88:106062. https://doi.org/10.1016/j.asoc.2020.106062.

[67] Alam MN, Chakrabarti S, Sharma A, Srivastava SC. An adaptive protection Scheme for AC microgrids using μPMU based topology processor. In: 2019 IEEE international conference on environment and electrical engineering and 2019 IEEE industrial and commercial power systems Europe (EEEIC / I&CPS Europe). IEEE; 2019. p. 1–6. https://doi.org/10.1109/EEEIC.2019.8783396.

[68] Ferreira RR, Colorado PJ, Grilo AP, Teixeira JC, Santos RC. Method for identification of grid operating conditions for adaptive overcurrent protection during intentional islanding operation. Int J Electr Power Energy Syst 2019;105:632–41. https://doi.org/10.1016/j.ijepes.2018.09.004.

[69] Ojaghi Vahid MM. Use of clustering to reduce the number of different setting groups for adaptive coordination of overcurrent relays. IEEE Trans Power Deliv 2018;33:1204–12. https://doi.org/10.1109/tpwrd.2017.2749321.

[70] Kazem Zanjani, Kamwa Innocent MGMM. Application of μPMUs for adaptive protection of overcurrent relays in microgrids. IET Gener Transm Distrib 2018;12:4061–8. https://doi.org/10.1049/iet-gtd.2018.5898.

[71] Mohammad Fani, Karami-Horestani Alireza BD. Adaptive protection coordination scheme against the staircase fault current waveforms in PV-dominated distribution systems. IET Gener Transm Distrib 2018;12:2065–71. https://doi.org/10.1049/iet-gtd.2017.0586.

[72] Atteya AI, El Zonkoly AM, Hamdy A. Ashour. Optimal relay coordination of an adaptive protection scheme using modified PSO algorithm. In: 2017 Nineteenth international Middle East power systems conference (MEPCON). IEEE; 2017. p. 689–94. https://doi.org/10.1109/MEPCON.2017.8301256.

[73] Papaspiliotopoulos George N, Vasilis Kleftakis, VAK Hatziargyriou Nikos. Hardware-in-the-loop design and optimal setting of adaptive protection schemes for distribution systems with Distributed generation. IEEE Trans Power Deliv 2017;32:393–400. https://doi.org/10.1109/tpwrd.2015.2509784.

[74] El Naily N, SaadM Saad, Hussein T, Mohamed FA. Minimizing the impact of distributed generation of a weak distribution network with an artificial intelligence technique. Appl Sol Energy 2017;53:109–22. https://doi.org/10.3103/S0003701X17020128.

[75] Muda Premalata HJ. Superimposed adaptive sequence current based microgrid protection: A new technique. IEEE Trans Power Deliv 2017;32:757–67. https://doi.org/10.1109/tpwrd.2016.2601921.

[76] Zhao Q, Liu S. Relay protection based on Multi-Agent System. In: 2011 International conference on electrical and control engineering. IEEE; 2011. p. 286–9. https://doi.org/10.1109/ICECENG.2011.6057028.

[77] de Azevedo R, Cintuglu MH, Ma T, Mohammed OA. Multiagent-based optimal microgrid control using fully distributed diffusion strategy. IEEE Trans Smart Grid 2017;8:1997–2008. https://doi.org/10.1109/TSG.2016.2587741.

[78] Cisneros-Saldana JID, Samal S, Begovic MM, Samantaray SR. On protection schemes for AC microgrids: challenges and opportunities. IEEE Trans Ind Appl 2024;60:4843–54. https://doi.org/10.1109/TIA.2024.3353721.

[79] Sarangi S, Sahu BK, Rout PK. Distributed generation hybrid AC/DC microgrid protection: A critical review on issues, strategies, and future directions. Int J Energy Res 2020;44:3347–64. https://doi.org/10.1002/er.5128.

[80] Liu Z, Su C, Hoidalen HK, Chen Z. A multiagent system-based protection and control scheme for distribution system with distributed-generation integration. IEEE Trans Power Deliv 2017;32:536–45. https://doi.org/10.1109/TPWRD.2016.2585579.

[81] Rahman MS, Isherwood N, Oo AMT. Multi-agent based coordinated protection systems for distribution feeder fault diagnosis and reconfiguration. Int J Electr Power Energy Syst 2018;97:106–19. https://doi.org/10.1016/j.ijepes.2017.10.031.

[82] Shirazi E, Jadid S. A multiagent design for self-healing in electric power distribution systems. Electr Power Syst Res 2019;171:230–9. https://doi.org/10.1016/j.epsr.2019.02.025.

[83] Nasr M-A, Hooshyar A. Power swing in systems with inverter-based resources—part II: impact on protection systems. IEEE Trans Power Deliv 2024;39:1903–17. https://doi.org/10.1109/TPWRD.2024.3382843.

[84] Aghdam TS, Kazemi Karegar H, Zeineldin HH. Variable tripping time differential protection for microgrids considering DG stability. IEEE Trans Smart Grid 2019;10:2407–15. https://doi.org/10.1109/TSG.2018.2797367.

[85] Abbaspour E, Fani B, Heydarian-Forushani E. A bi-level multi agent based protection scheme for distribution networks with distributed generation. Int J Electr Power Energy Syst 2019;112:209–20. https://doi.org/10.1016/j.ijepes.2019.05.001.

[86] Satuyeva B, Sultankulov B, Nunna HSVSK, Kalakova A, Doolla S. Q-learning based protection Scheme for microgrid using Multi-Agent System. In: 2019 international conference on smart energy systems and technologies (SEST). IEEE; 2019. p. 1–6. https://doi.org/10.1109/SEST.2019.8849088.

[87] Uzair M, Li L, Zhu JG, Eskandari M. A protection scheme for AC microgrids based on multi-agent system combined with machine learning. In: 2019 29th Australasian universities power engineering conference (AUPEC). IEEE; 2019. p. 1–6. https://doi.org/10.1109/AUPEC48547.2019.211845.

[88] Abbaspour E, Fani B, Karami-Horestani A. Adaptive scheme protecting renewable-dominated micro-grids against usual topology-change events. IET Renew Power Gener 2021:15. https://doi.org/10.1049/rpg2.12193.

[89] Sampaio FC, Leão RPS, Sampaio RF, Melo LS, Barroso GC. A multi-agent-based integrated self-healing and adaptive protection system for power distribution systems with distributed generation. Electr Power Syst Res 2020;188:106525. https://doi.org/10.1016/j.epsr.2020.106525.

[90] Ataei MA, Gitizadeh M. A distributed adaptive protection scheme based on multi-agent system for distribution networks in the presence of distributed generations. IET Gener Transm Distrib 2022;16:1521–40. https://doi.org/10.1049/gtd2.12351.

[91] Daryani MJ, Karkevandi AE. Decentralized cooperative protection strategy for smart distribution grid using Multi-Agent System. In: 2018 6th international Istanbul smart grids and cities congress and fair (ICSG). IEEE; 2018. p. 134–8. https://doi.org/10.1109/SGCF.2018.8408958.

[92] Daryani MJ, Karkevandi AE, Usta O. Multi-agent approach to wide-area integrated adaptive protection system of microgrid for pre-and post-contingency conditions. In: Proceedings - 2018 IEEE PES innovative smart grid technologies conference Europe, ISGT-Europe 2018; 2018. https://doi.org/10.1109/ISGTEurope.2018.8571785.

[93] Kiani A, Fani B, Shahgholian G. A multi-agent solution to multi-thread protection of DG-dominated distribution networks. Int J Electr Power Energy Syst 2021;130:106921. https://doi.org/10.1016/J.IJEPES.2021.106921.

[94] Dizioli FAS, Barra PHA, Menezes TS, Lacerda VA, Coury DV, Fernandes RAS. Multi-agent system-based microgrid protection using angular variation: an embedded approach. Electr Power Syst Res 2023;220:109324. https://doi.org/10.1016/j.epsr.2023.109324.

[95] De La Cruz J, Vasquez JC, Guerrero JM, Luna EG, Candelo-Becerra JE. Adaptive multi-agent-zonal protection scheme for AC microgrids. In: 2023 25th European conference on power electronics and applications (EPE'23 ECCE Europe). IEEE; 2023. p. 1–9. https://doi.org/10.23919/EPE23ECCEEurope58414.2023.10264617.

[96] Alzahrani S, Sinjari K, Mitra J. Multi-agent and state observer-based technique for microgrid protection. IEEE Trans Ind Appl 2024;60:2697–705. https://doi.org/10.1109/TIA.2023.3336632.

[97] Waqar H, Bukhari SBA, Wadood A, Albalawi H, Mehmood KK. Fault identification, classification, and localization in microgrids using superimposed components and Wigner distribution function. Front Energy Res 2024:12. https://doi.org/10.3389/fenrg.2024.1379475.

[98] Do Nascimento LL, Rolim JG. Multi-agent system for adaptive protection in microgrids. In: 2013 IEEE PES conference on innovative smart grid technologies, ISGT LA 2013; 2013. https://doi.org/10.1109/ISGT-LA.2013.6554435.

[99] Ma Z, Xiao M, Xiao Y, Pang Z, Poor HV, Vucetic B. High-reliability and low-latency wireless communication for internet of things: challenges, fundamentals, and enabling technologies. IEEE Internet Things J 2019;6:7946–70. https://doi.org/10.1109/JIOT.2019.2907245.

[100] Marzal S, Salas R, González-Medina R, Garcerá G, Figueres E. Current challenges and future trends in the field of communication architectures for microgrids. Renew Sust Energ Rev 2018;82:3610–22. https://doi.org/10.1016/J.RSER.2017.10.101.

[101] Rath S, Pal D, Sharma PS, Panigrahi BK. A cyber-secure distributed control architecture for autonomous AC microgrid. IEEE Syst J 2020;15:3324–35. https://doi.org/10.1109/JSYST.2020.3020968.

[102] Zhou TL, Xiahou KS, Zhang LL, Wu QH. Multi-agent-based hierarchical detection and mitigation of cyber attacks in power systems. Int J Electr Power Energy Syst 2021;125:106516. https://doi.org/10.1016/j.ijepes.2020.106516.
[103] Bretas AS, Bretas NG, Carvalho BEB. Further contributions to smart grids cyber-physical security as a malicious data attack: proof and properties of the parameter error spreading out to the measurements and a relaxed correction model. Int J Electr Power Energy Syst 2019;104:43–51. https://doi.org/10.1016/j.ijepes.2018.06.039.
[104] Kondoro A, Ben Dhaou I, Tenhunen H, Mvungi N. A low latency secure communication architecture for microgrid control. Energies (Basel) 2021:14. https://doi.org/10.3390/en14196262.
[105] Zhang B, Dou C, Yue D, Zhang Z. Response hierarchical control strategy of communication data disturbance in micro-grid under the concept of cyber physical system. IET Gener Transm Distrib 2018;12:5867–78. https://doi.org/10.1049/iet-gtd.2018.5356.
[106] Memon AA, Kauhaniemi K. Real-time hardware-in-the-loop testing of IEC 61850 GOOSE-based logically selective adaptive protection of AC microgrid. IEEE Access 2021;9:154612–39. https://doi.org/10.1109/ACCESS.2021.3128370.
[107] Ghorbani J, Choudhry MA, Feliachi A. Real-time multi agent system modeling for fault detection in power distribution systems. In: 2012 North American power symposium (NAPS). IEEE; 2012. p. 1–6. https://doi.org/10.1109/NAPS.2012.6336398.
[108] Ghorbani MJ, Mohammadi FD, Choudhry MA, Feliachi A, Ashby DA. Hardware design for distributed MAS-based fault location in power distribution systems. In: IEEE Power and Energy Society General Meeting. 2014- Octob. IEEE; 2014. p. 1–5. https://doi.org/10.1109/PESGM.2014.6938829.
[109] Wang J, Pratt A, Prabakar K, Miller B, Symko-Davies M. Development of an integrated platform for hardware-in-the-loop evaluation of microgrids prior to site commissioning. Appl Energy 2021;290:116755. https://doi.org/10.1016/j.apenergy.2021.116755.
[110] Mobashsher MM, Abdoos AA, Hosseini SM, Hashemi SM, Sanaye-Pasand M. An accelerated distance protection scheme for the lines connected to inverter-based resources. IEEE Syst J 2023;17:6272–81. https://doi.org/10.1109/JSYST.2023.3281345.
[111] Afshari A, Karrari M, Baghaee HR, Gharehpetian GB. Resilient cooperative control of AC microgrids considering relative state-dependent noises and communication time-delays. IET Renew Power Gener 2020;14:1321–31. https://doi.org/10.1049/iet-rpg.2019.1180.
[112] Li Z, Chen G. Distributed adaptive control Scheme for islanded AC microgrids with tolerance to uncertain communication links. IEEE Syst J 2022;16:2741–52. https://doi.org/10.1109/JSYST.2021.3128737.
[113] Dou C, Yue D, Guerrero JM, Xie X, Hu S. Multiagent system-based distributed coordinated control for radial DC microgrid considering transmission time delays. IEEE Trans Smart Grid 2017;8:2370–81. https://doi.org/10.1109/TSG.2016.2524688.
[114] Ashraf S, Hasan O, Evkay I, Selamogullari US, Baysal M. Recent trends and developments in protection systems for microgrids incorporating distributed generation. WIREs. Energy Environt 2024:13. https://doi.org/10.1002/wene.532.
[115] dos Reis FB, Pinto JOCP, dos Reis FS, Issicaba D, Rolim JG. Multi-agent dual strategy based adaptive protection for microgrids. Sustain Energy, Grids Networks 2021;27:100501. https://doi.org/10.1016/j.segan.2021.100501.
[116] Khazaei AA, Mahmoudi A. Decentralized adaptive protection structure for microgrids based on multi–agent systems. In: 2019 Iranian Conference on Renewable Energy & Distributed Generation (ICREDG). IEEE; 2019. p. 1–8. https://doi.org/10.1109/ICREDG47187.2019.194143.
[117] Faria IM, Furlan RH, Martins PET, Menezes TS, Oleskovicz M, Coury DV. The proposition of a multiagent system for adaptive protection of a distribution system. In: 2018 Simposio Brasileiro de Sistemas Eletricos (SBSE). IEEE; 2018. p. 1–6. https://doi.org/10.1109/SBSE.2018.8395648.
[118] Tazi K, Abbou FM, Abdi F. Multi-agent system for microgrids: design, optimization and performance. Artif Intell Rev 2020;53:1233–92. https://doi.org/10.1007/s10462-019-09695-7.
[119] Deng C, Wen C, Zou Y, Wang W, Li X. A hierarchical security control framework of nonlinear CPSs against DoS attacks with application to power sharing of AC microgrids. IEEE Trans Cybern 2022;52:5255–66. https://doi.org/10.1109/TCYB.2020.3029045.